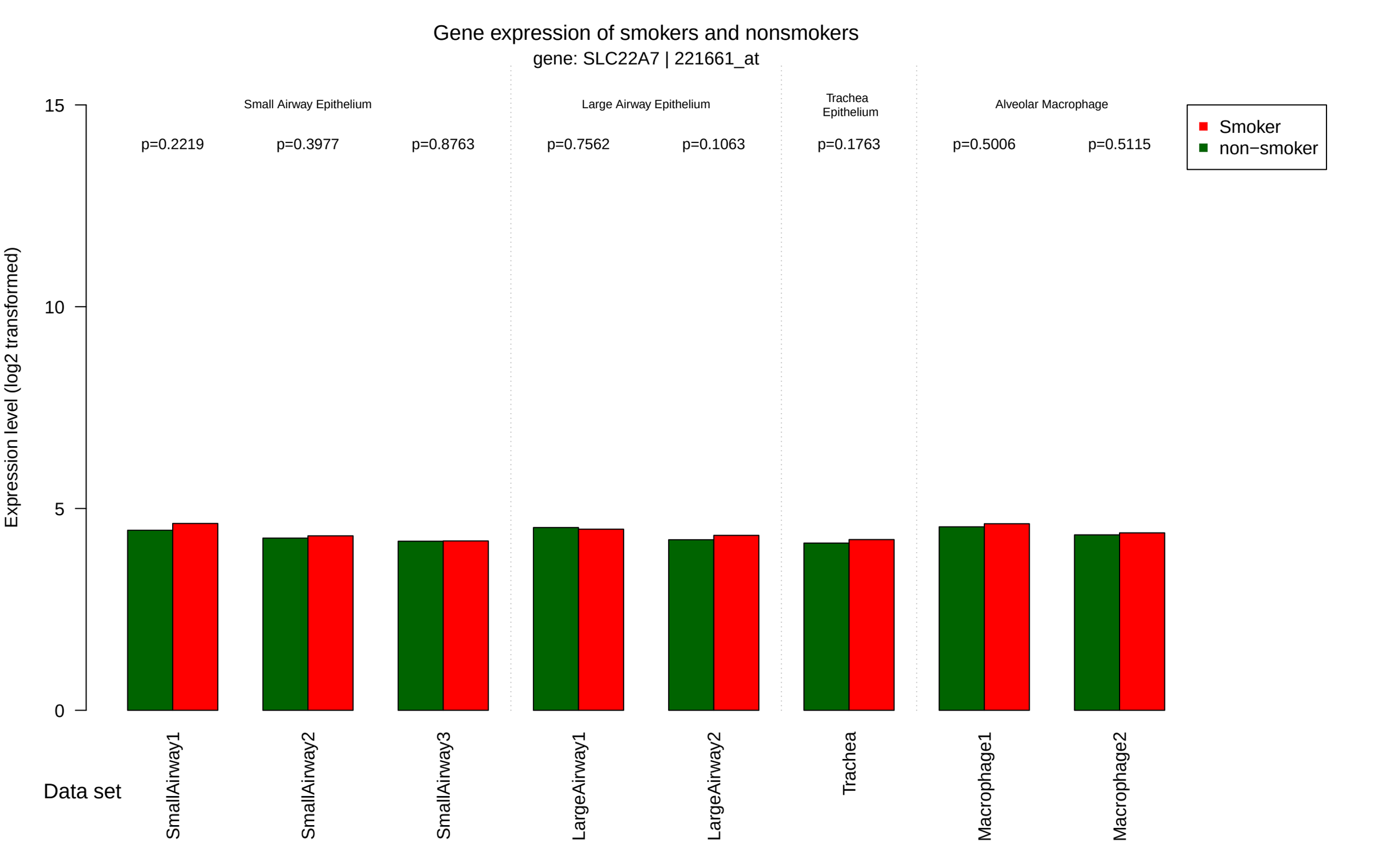
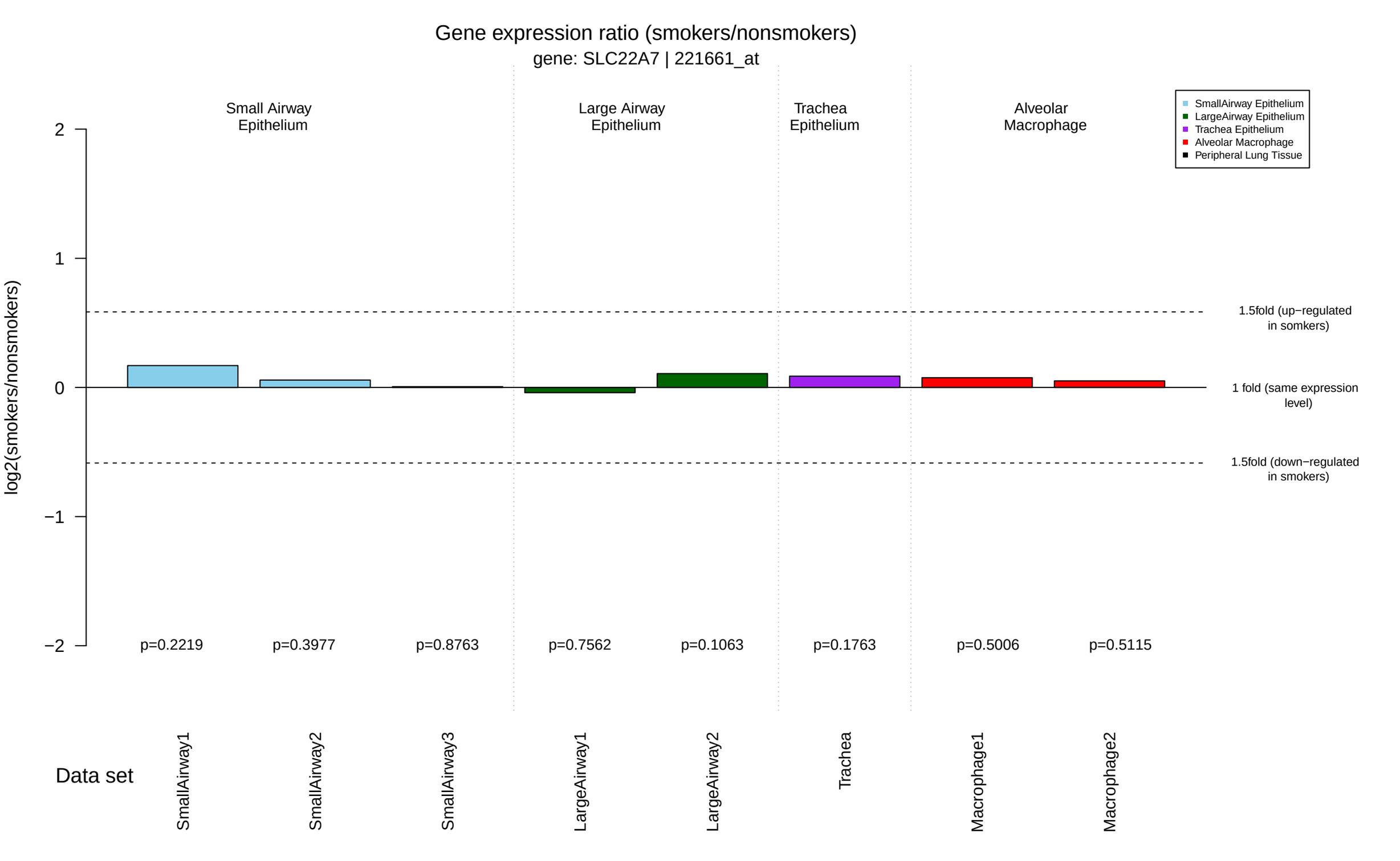
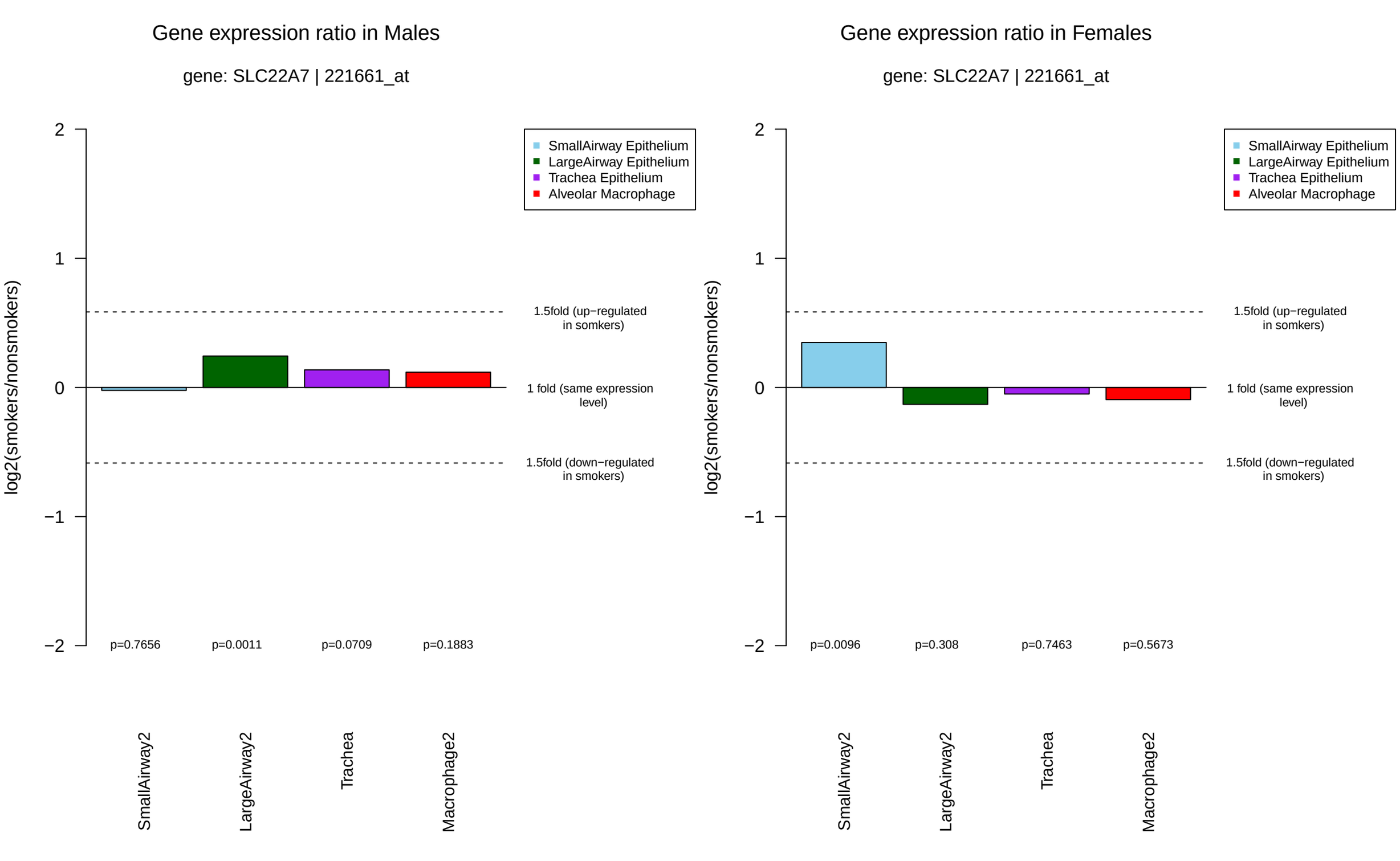
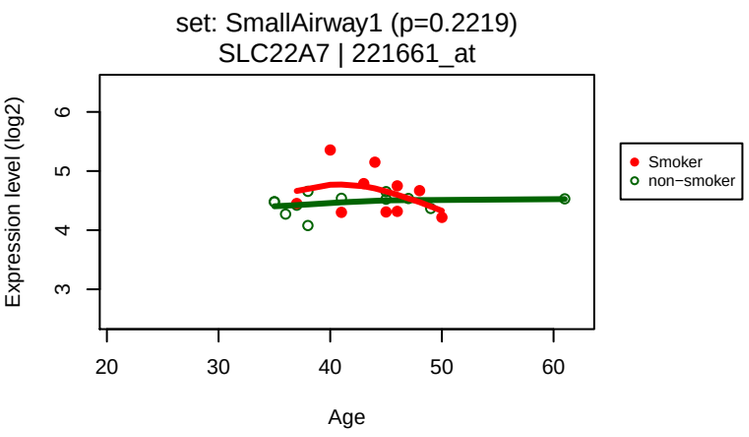
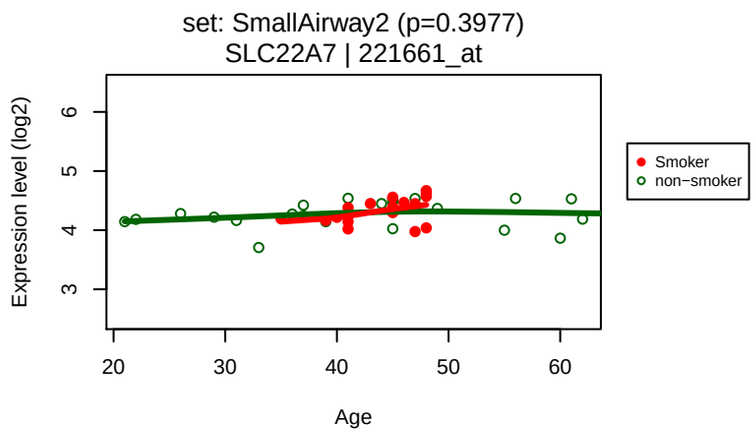
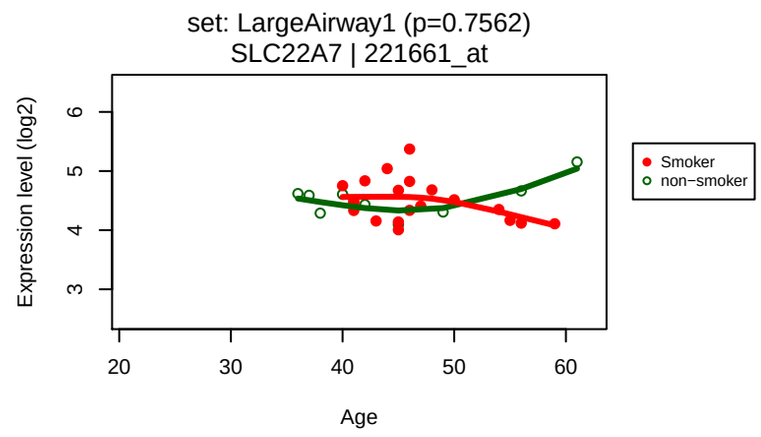

Gene expression of smokers and nonsmokers
gene: SLC22A7 | 221661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2219
p=0.3977
p=0.8763
p=0.7562
p=0.1063
p=0.1763
p=0.5006
p=0.5115
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

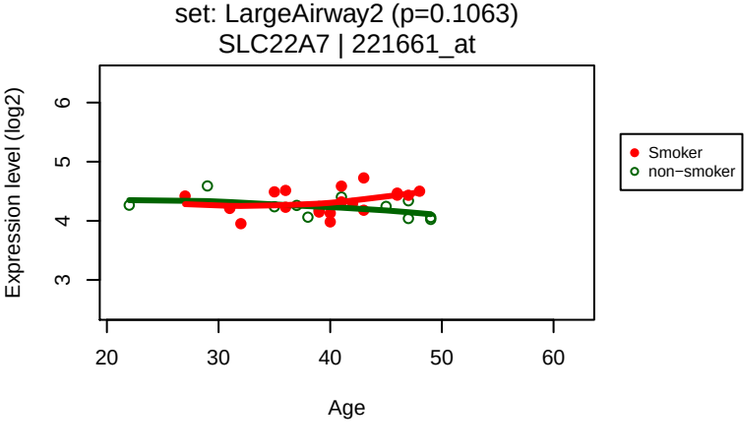

Gene expression ratio (smokers/nonsmokers)
gene: SLC22A7 | 221661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2219
p=0.3977
p=0.8763
p=0.7562
p=0.1063
p=0.1763
p=0.5006
p=0.5115
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

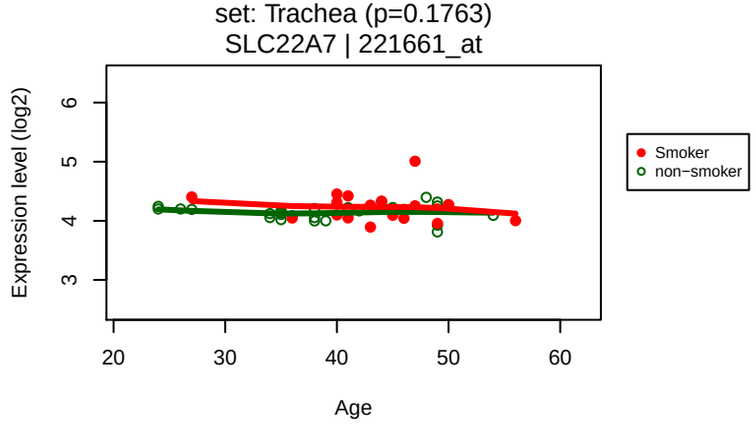

Gene expression ratio in Males
gene: SLC22A7 | 221661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7656
p=0.0011
p=0.0709
p=0.1883
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC22A7 | 221661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0096
p=0.308
p=0.7463
p=0.5673
SmallAirway2
LargeAirway2
Trachea
Macrophage2

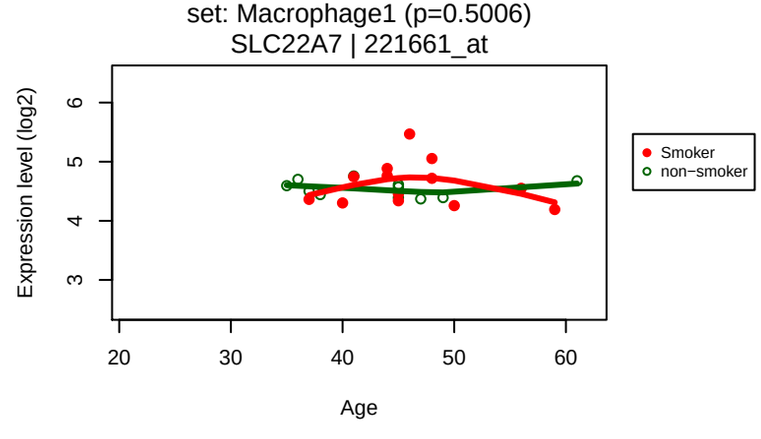

set: SmallAirway1 (p=0.2219)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker

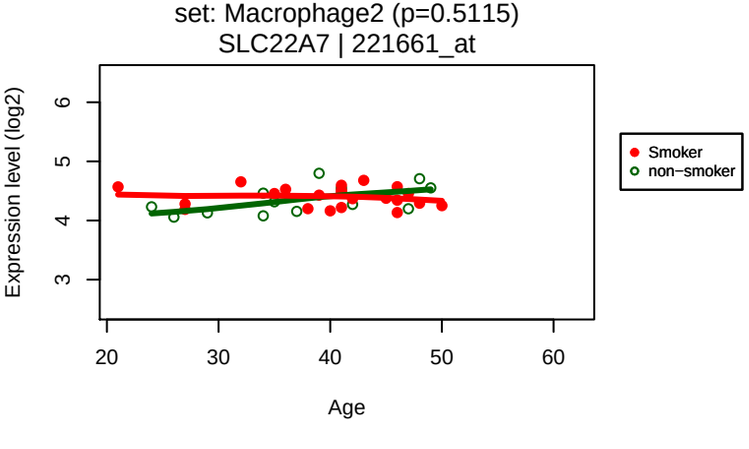

set: SmallAirway2 (p=0.3977)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker

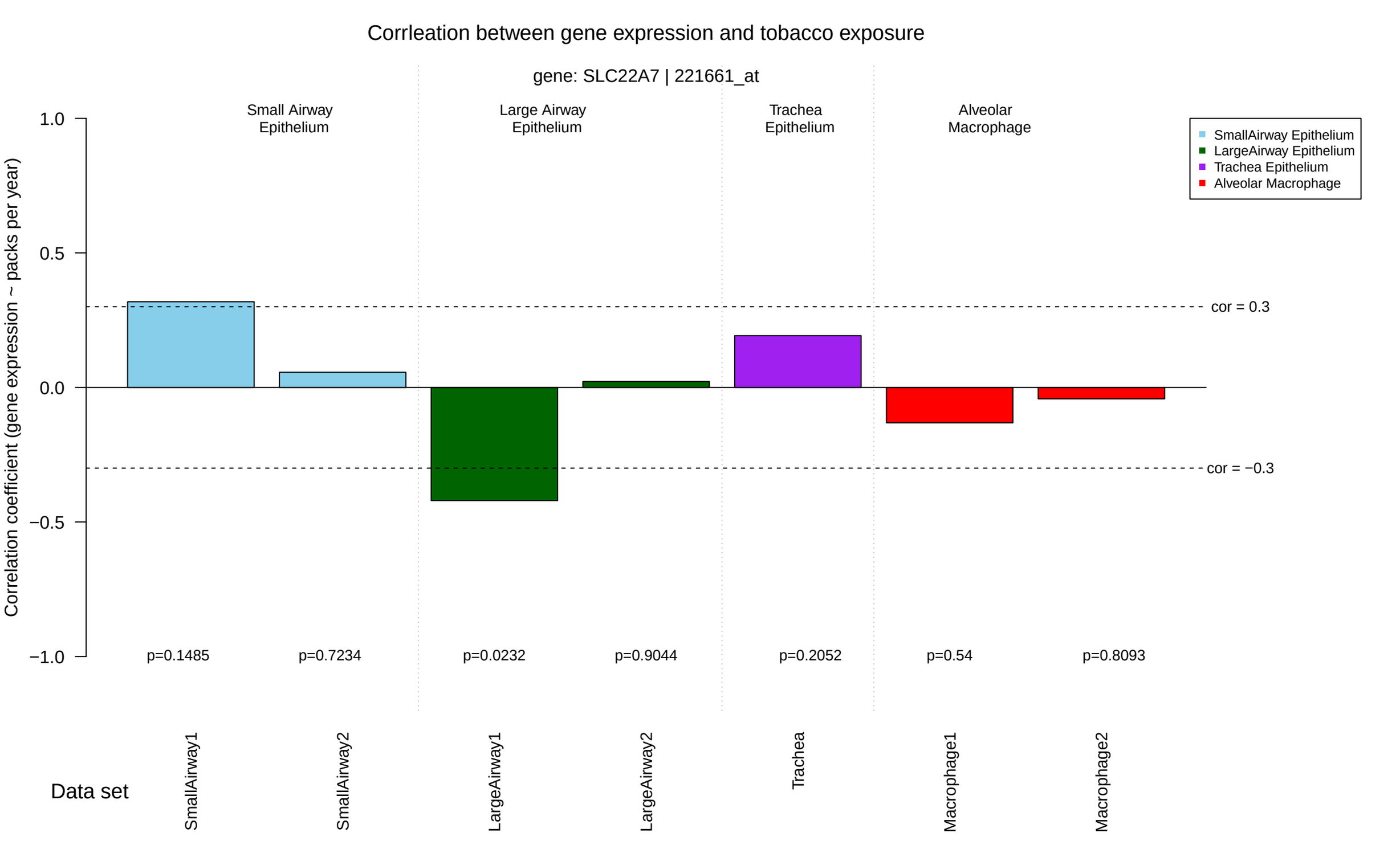

set: LargeAirway1 (p=0.7562)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1063)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1763)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5006)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5115)
SLC22A7 | 221661_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC22A7 | 221661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1485
p=0.7234
p=0.0232
p=0.9044
p=0.2052
p=0.54
p=0.8093
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2